Introduction/Motivation: In this project, we analyzed a set of 252 men to develop a rule-of-thumb for estimating body fat percentage using predictors such as abdominal circumference and weight. Measuring body fat percentage often requires specialized and inconvenient methods, such as underwater weighing for density measurements, that are not practical in everyday settings. Therefore, we need to use simple predictors such as abdominal circumference and weight to determine body fat percentage. Our goal for this project is to identify a model that is both accurate and easy to use, while also providing meaningful insights into how certain measurements of the body relate to body fat.

Background Information/Data Cleaning:
The mean body fat of the men in the data is 18.9% (SD = 7.6%). The values in our data span from athletic levels (2%) to clinical obesity (45%). This wide spread allows us to produce a stronger model. It suggests the model we used was trained on a diverse group of body compositions, not just a narrow "average" group. The primary predictors of our model were abdomen circumference, with a mean of 92.6 cm and a standard deviation of 10.7, and weight, where the mean was 179.1 lbs with a standard deviation of 29.2 lbs. The abdomen measurements have the highest variability among all circumference measurements. This high variance makes it the most "informative" variable for predicting diverse body fat percentages. While the average man in the study weighs 179 lbs, the data includes individuals up to 363 lbs. This diversity allows the model to be effective in predicting a wider range of body fat percentages. For our first case, we removed an individual with a body fat % 0 because it is physically impossible. For our second case, we removed a 205 lb person who is under 3 feet tall because this is also biologically impossible

Motivation for Model/Choosing Model/Final Model/Rule of Thumb
Our final model is Body Fat % ≈ (0.91 × Abdomen) - (0.14 × Weight) - 40.7. As an example, a man weighing 200 lbs with an abdomen circumference of 100 cm is expected to have a body fat percentage of 22% based on our model. He has a 95% chance of having a body fat percentage between 14.7% and 31.0%, assuming our model is accurate. When we interpreted our model, our estimated coefficients were 0.91 and -0.14, which are in the units of body fat percentage per abdomen circumference in centimeters and body fat percentage per pound of body weight, respectively. This means that for every centimeter increase in abdomen circumference (holding weight constant), the model predicts that body fat percentage will increase, on average, by a factor of 0.91. For every one-pound increase in weight (holding abdomen circumference constant), the model predicts that body fat percentage will decrease, on average, by 14%. To gain this conclusion, we first found the single most relevant predictor, Abdomen, with an adjusted R^2 of 0.653 and a T score of 21.67, the largest among the rest of the predictors. We then attempted to add a second predictor, turning our SLR model into an MLR model, in which we found that weight was the most significant predictor among the rest, with a T score of -7.122 and an adjusted R^2 that explained a proportion of 0.711 of the variability of the model (R^2). This increase of 5.8 percentage points was enough to justify the added complexity of the second predictor. We then attempted to run a third predictor through the model: wrist. Among the predictors, wrist was the only one with a T-score that resulted in a p-value below the 0.05 threshold, at 0.007. Although significant, the three-predictor model only had an adjusted R^2 of 0.718, only 0.7 percentage points larger than the prior model. We decided that this increase was not worth the added complexity of the new predictor. After finding the model, we hypothesized that there was an interaction between abdomen and weight, in which abdomen:weight had a significant t value of -3.179 and an adjusted R^2 of 0.721, which, like the previous models, the 1 percentage point increase did not justify the complex math. Mallows' Cp suggested that 7 predictor variables may improve fit, but our two-predictor model resulted in a BIC score of 1435 vs the complex model, resulting in a BIC score of 1444, indicating that the improved fit did not justify the added complexity. Although ANOVA tests indicated that each of these models resulted in statistically significant improvements in fit at the alpha = 0.05 level, these improvements were marginal. We chose the two predictor model as its coefficients were statistically significant, it is easily interpretable, and it does not have unnecessary complexity.

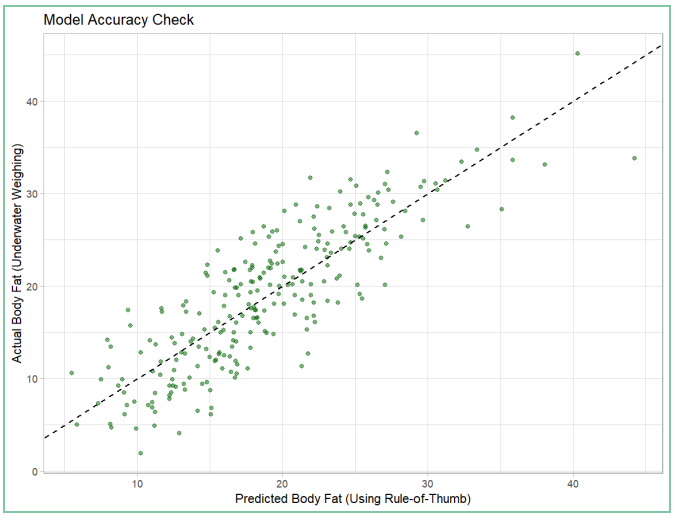


Figure 1

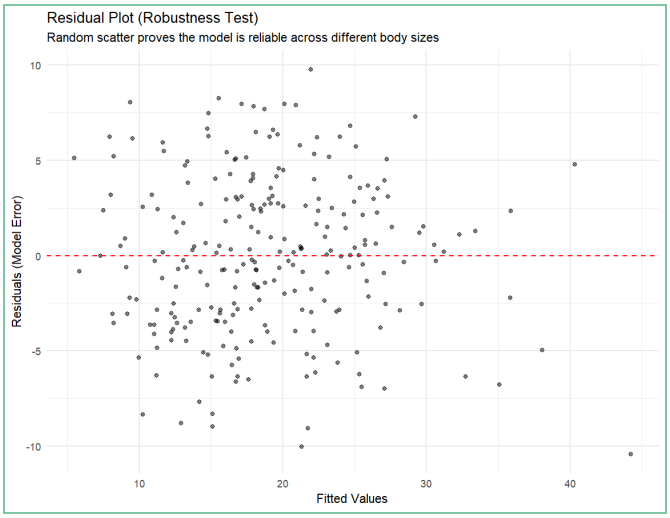


Figure 2

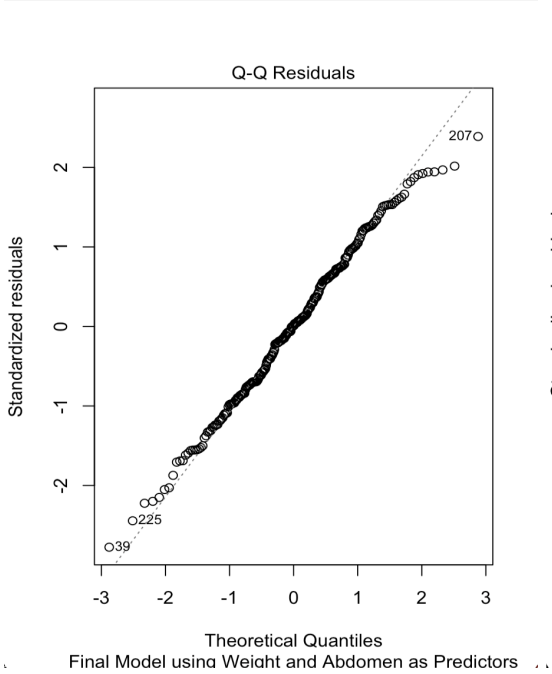
Statistical Analysis/Hypothesis Testing/Inference:

We conducted hypothesis tests to see whether abdomen circumference and weight are significant in predicting the outcome. For each predictor, we tested the null hypothesis of H0: βabdomen = 0 and H0: βweight = 0 against the alternative hypotheses that the coefficients are not 0. Using the t-tests from the regression output, the abdomen circumference had a t-value of 17.336 with a p-value smaller than 2 x 10^-16, while the weight had a t-value of -7.122 with a p-value of 1.16 x 10^-11. Because both p-values are less than the significance level of 0.05, we reject both null hypotheses. The data indicates that both abdomen circumference and weight are both statistically significant predictors of body fat percentage, when controlling for each other. Next, we found our R^2 to be 0.711, which implies that approximately 71.1% of the variability in body fat percentage can be explained by abdomen circumference and weight. This means that our model does a solid job of predicting body fat using these variables. Finally, the estimated slopes for abdomen circumference and weight, respectively, are 0.909 and -.137, and the intercept is -40.7. This comes from the estimated regression equation of: Body Fat = -40.7 + 0.909(Abdomen) - 0.137(Weight). Using 95% confidence intervals, the abdomen circumference falls between 0.805 and 1.012, while the weight falls between -.175 and -.099. These results imply that on average, for each increase of 1 cm in abdominal circumference, body fat increases by a factor of 0.909, holding weight constant. Additionally, for each 1 lb. increase in weight, body fat percentage decreases by a factor of 0.137, holding abdominal circumference constant. Since neither confidence interval includes 0, we reject the hypotheses that the coefficients are equal to 0. In simple terms, both abdomen circumference and weight have statistically significant effects on body fat percentage and are important predictors in the model. For our final model, we decided to round these coefficients to 0.91 (abdomen) and -0.14 (weight)

Model Diagnostics:

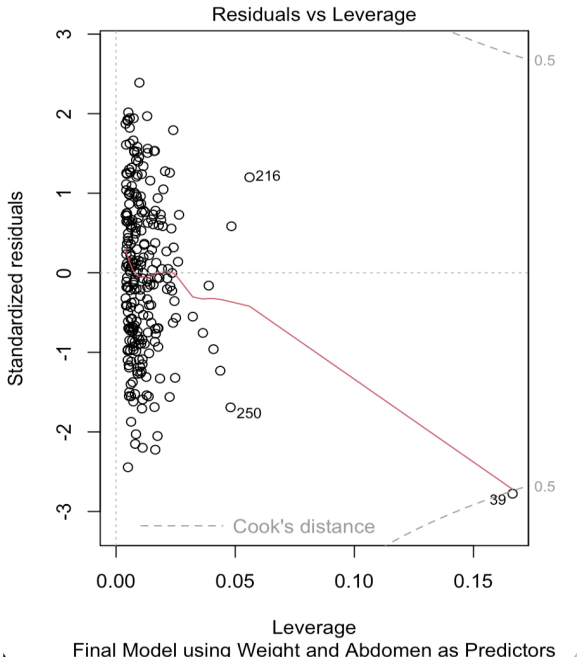
We checked the following four assumptions for MLR. First, we checked Linearity using a residual vs fitted plot (see Figure 2). Because there was no leftover pattern, we believed linearity is plausible, even though there is a slight violation of linearity as the LOESS line trended downwards as the fitted values increased (This is likely due to the high leverage point of observation 39). Second, we checked Heteroskedasticity using the same residual vs fitted plot (see Figure 2). The residuals maintain a relatively even spread throughout the model, showing no clear evidence of heteroskedasticity. Third, we checked for normality using a QQ Plot (see Figure 3). Since the residuals approximately fell on the 45-degree line, the normality assumption was reasonably satisfied. Finally, we checked for Collinearity by checking the VIF scores. Though each of the predictors had a VIF score of 4.6, hinting at moderate multicollinearity, both variables remain statistically significant, and the adjusted R^2 is strong (0.711), meaning the performance of the model is not affected. There were three outliers in our data: observations 39, 216, and 250. When reviewing our Residuals vs Leverage Plot (See figure 4), we noticed that observations 250 and 216 have moderate leverage, but observation 39 has a Cook's score of approximately 0.5, indicating a high leverage point. After reviewing the data of the observation, we noticed that

observation 39 had an abdomen circumference of 148.1 cm, a weight of 363.15 lbs, but only a body fat percentage of 33.8%. While this was confusing at first glance, we quickly realized that observation 39 may be a strongman with high muscle mass, which is responsible for his heavy weight and large abdomen circumference, while having a low body fat percentage. Even though his data may skew our results due to the high leverage, keeping this point helps the model better understand extreme body types, even if it makes the overall average prediction slightly less perfect.



Final Model using Weight and Abdomen as Predictors

Figure 3



Final Model using Weight and Abdomen as Predictors

Figure 4

Model Strengths/Weaknesses:

Some strengths of our model include simplicity, strong predictive accuracy, and a correction for lean muscle mass via the negative weight coefficient. In particular, our model satisfies the linear regression assumptions of normality and homoscedasticity, bringing credence to our results/interpretation in providing a reliable rule-of-thumb for body fat %. Some weaknesses of our model include the exclusion of females, a standard error of approximately 4%, and a reliance on linear relationships that may not hold at extremes. In particular, for certain groups of males, such as strongmen or those with obesity, the model may provide underestimates of fat and not be as reliable because of muscle-to-fat ratios that differ from those of the more average males used to train our model.

Conclusion/Discussion:

In this project, we successfully created a simple way for men to estimate their body fat percentage without using expensive lab equipment. By using only weight and abdomen size, the model provides a reliable "rule of thumb" that is easy for anyone to use. We chose this specific model because it balances accuracy with simplicity. While it isn't perfect, it explains about 71.1% of the differences in body fat among most men. Ultimately, we have shown that basic body measurements can give a solid, practical look at a person's health.